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Systems and methods for multi-accessory mount assembly for a firearm

Abstract

A multi-accessory mount assembly for use with a firearm, and methods for coupling the assembly to the firearm are provided. The assembly may include a proximal portion having a lower surface configured to be removably coupled to the firearm and an upper surface configured to be removably coupled to a sub mount. The sub mount may be configured to be removably coupled to a first firearm accessory, e.g., an optic scope, a red dot sight, a reflex sight, a night vision monocular or scope, or a magnifier. The assembly further may include a distal portion extending distally from the proximal portion via a neck portion, such that the distal portion is elevated higher than the proximal portion. An upper surface of the distal portion may be configured to be removably coupled to a second firearm accessory, e.g., an infrared laser or a night vision monocular or scope.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5134798	12/1991	Lee	42/114	F41G 11/003
5941489	12/1998	Fanelli et al.	N/A	N/A
6854206	12/2004	Oz	N/A	N/A
6895708	12/2004	Kim et al.	N/A	N/A
7240451	12/2006	Blair	N/A	N/A
D594925	12/2008	Moody et al.	N/A	N/A
7765730	12/2009	Da Keng	N/A	N/A
D632753	12/2010	Leighton et al.	N/A	N/A
D654134	12/2011	Edge et al.	N/A	N/A
8146283	12/2011	Genes et al.	N/A	N/A
8240075	12/2011	Mullin	N/A	N/A
D673239	12/2011	Cheng	N/A	N/A
D713921	12/2013	Williams et al.	N/A	N/A
RE45185	12/2013	Stanowski	N/A	N/A
D715393	12/2013	Williams et al.	N/A	N/A
8857096	12/2013	McClintock	N/A	N/A
8935875	12/2014	Collin	42/124	F41G 11/003
D726864	12/2014	Huang, Sr.	N/A	N/A
9015982	12/2014	Powers	N/A	N/A
D732631	12/2014	Cheng et al.	N/A	N/A
9057584	12/2014	Chung	N/A	N/A
D738986	12/2014	Anderson	N/A	N/A
9157696	12/2014	Dextraze	N/A	N/A
D761929	12/2015	Hormann	N/A	N/A
9441915	12/2015	Zimmer	N/A	N/A
9470480	12/2015	Kirchhoff	N/A	N/A
9488446	12/2015	Zimmer	N/A	N/A
9523557	12/2015	Sharron et al.	N/A	N/A
9523558	12/2015	Visinski et al.	N/A	N/A
9574852	12/2016	McKillips et al.	N/A	N/A
D784481	12/2016	Ding et al.	N/A	N/A

D787007	12/2016	Maughn	N/A	N/A
9689646	12/2016	Story et al.	N/A	N/A
9784527	12/2016	Helms	N/A	N/A
9857142	12/2017	Sharron et al.	N/A	N/A
D820377	12/2017	Boyd et al.	N/A	N/A
D822144	12/2017	Geissele	N/A	N/A
D824481	12/2017	Geissele	N/A	N/A
10024632	12/2017	Oglesby	N/A	N/A
10036614	12/2017	Ruiz	N/A	N/A
D825022	12/2017	Geissele et al.	N/A	N/A
10060705	12/2017	Compton et al.	N/A	N/A
D831778	12/2017	Geissele et al.	N/A	N/A
D831780	12/2017	Geissele	N/A	N/A
10088277	12/2017	Zimmer	N/A	N/A
10101126	12/2017	Sharron et al.	N/A	N/A
10101127	12/2017	Visinski et al.	N/A	N/A
10119787	12/2017	Geissele	N/A	N/A
10151557	12/2017	Sharron et al.	N/A	N/A
D837330	12/2018	Ding et al.	N/A	N/A
10352656	12/2018	Ding et al.	N/A	N/A
10359258	12/2018	Geissele	N/A	N/A
D877843	12/2019	Yu	N/A	N/A
10634453	12/2019	Spuhr	N/A	N/A
10670374	12/2019	Storch	N/A	N/A
10845162	12/2019	Geissele	N/A	N/A
10871344	12/2019	Brown, Jr.	N/A	N/A
10921090	12/2020	Schatz et al.	N/A	N/A
11150049	12/2020	Kennair, Jr.	N/A	F41G 11/003
11680771	12/2022	Kielsmeier et al.	N/A	N/A
11680776	12/2022	Nielsen	N/A	N/A
2005/0229465	12/2004	Dextraze	N/A	N/A
2006/0059762	12/2005	Blair	N/A	N/A
2007/0199225	12/2006	Haugen	N/A	N/A
2011/0000119	12/2010	Desomma et al.	N/A	N/A
2012/0017482	12/2011	Chvala	N/A	N/A
2012/0311909	12/2011	Cheng	N/A	N/A
2013/0118051	12/2012	Baker	N/A	N/A
2014/0068994	12/2013	Schultz et al.	N/A	N/A
2014/0259854	12/2013	Williams et al.	N/A	N/A
2015/0020429	12/2014	Savoy et al.	N/A	N/A
2015/0253104	12/2014	Holland	N/A	N/A
2016/0327371	12/2015	Teetzel et al.	N/A	N/A
2016/0363416	12/2015	Klotz	N/A	N/A
2017/0059280	12/2016	Fravor et al.	N/A	N/A
2017/0254615	12/2016	Kennair, Jr.	N/A	N/A
2017/0343315	12/2016	Beaty et al.	N/A	N/A
2018/0087875	12/2017	Yost	N/A	N/A
2018/0135946	12/2017	Geissele	N/A	N/A
2020/0200505	12/2019	Brown, Jr.	N/A	N/A
2021/0293512	12/2020	Duncan, III et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
211601731	12/2019	CN	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion mailed Jun. 27, 2023 issued in corresponding International Application No. PCT/US2022/047923 (6 pages). cited by applicant

GBRS Group Hydra Optic Mount Kit, Rainer Arms Online Product Page. Earliest Review Date: Nov. 14, 2022. Date Accessed: Apr. 1, 2024. Retrieved from Internet URL:

<https://www.rainierarms.com/gbrs-group-hydra-optic-mount-kit-eotech/> (Year: 2022). cited by applicant

EXPS3 Holographic Sight Riser Mount, Amazon Online Product Page. Earliest Publication Date: Feb. 16, 2023. Date Accessed: Apr. 1, 2024. Retrieved from Internet URL:

<https://www.amazon.com/rdzzwxw-Tactical-Holographic-Adapter-Picatinny/dp/BOBVZNJ7KD> (Year: 2023). cited by applicant

GBRS Group Hydra Mount Overview, YouTube-GBRS Group. Publication Date: Jul. 5, 2022. Date Accessed: Apr. 1, 2024. Retrieved from You Tube URL: <https://www.youtube.com/watch?v=WFMSEoXYmvE>

(Year: 2022). cited by applicant

Genuine GBRS Hydra Mount, YouTube-JLMK. Publication Date: Aug. 14, 2022. Date Accessed: Apr. 1, 2024. Retrieved from YouTube URL : <https://www.youtube.com/watch?v=QhQBLcixwk4>

(Year: 2022). cited by applicant

GBRS Group Hydra Mount Kit, GBRS Group Online Product Page. Earliest Publication Date: Jul. 2, 2023. Date Accessed: Apr. 1, 2024. Retrieved from Internet URL:

<https://gbrsgroupgear.com/products/gbrs-group-hydra-mount-kit> (Year: 2023). cited by applicant

<https://web.archive.org/web/20221008104219/https://www.airgunmarket.jp/SHOP/EVG-892B.html> (Oct. 8, 2022-Sep. 29, 2023). cited by applicant

<https://www.youtube.com/watch?app=desktop&v=oMLLKlNmRNc> (Jul. 23, 2022). cited by applicant

(<https://www.knightarmco.com/13809/shop/sighting/optic-mounts/reddot/aimpoint-micro-nvg-high-rise-mount-w-1913-rail-height-2-33>) (2022). cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is continuation of U.S. patent application Ser. No. 17/514,190, filed on Oct. 29, 2021 which claims the benefit of, U.S. Provisional Patent Application No. 63/170,791, filed Apr. 5, 2021, the entire contents of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The disclosure generally relates to a firearm and more particularly relates to systems and methods for a multi-accessory mount assembly for a firearm.

BACKGROUND

(2) Conventional optic scopes, magnifiers, and laser scopes have been used with firearms to facilitate aiming and to improve shooting accuracy. It is with respect to these and other considerations that the disclosure made herein is presented.

SUMMARY

(3) Some or all of the above needs and/or problems may be addressed by certain embodiments of the enhanced, elevated multi-accessory mount assembly for a firearm disclosed herein. The firearm may include a barrel with a muzzle end and a breech end. On the firearm usually positioned above the barrel and between the muzzle end and breech end is a mounting surface to which an optic may be mounted. Such mounting surfaces may include a Picatinny rail or a Weaver rail. According to certain embodiments, the multi-accessory mount assembly may include a proximal portion having a lower surface configured to be removably coupled to the firearm, for example, by attaching to a Picatinny or Weaver rail, and an upper surface configured to be removably coupled to an interchangeable sub mount. The interchangeable sub mount is configured to be removably coupled to one or more first firearm accessories, e.g., an optic scope, red dot sight, reflex sight, night vision monocular/scope, or magnifier. The assembly further may include a distal portion extending distally from the proximal portion via a neck portion, e.g., a goose neck, such that the distal portion is elevated higher than the proximal portion. An upper surface of the distal portion is configured to be removably coupled to a second firearm accessory, e.g., an infrared (IR) laser, or night vision monocular/scope.

(4) Other features and aspects of the multi-accessory mount assembly will be apparent or will become apparent to one with skill in the art upon examination of the following figures and the detailed description. All other features and aspects, as well as other systems, methods, and assembly embodiments, are intended to be included within the description and are intended to be within the scope of the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is set forth with reference to the accompanying drawings. The use of the same reference numerals may indicate similar or identical items. Various embodiments may utilize elements and/or components other than those illustrated in the drawings, and some elements and/or components may not be present in various embodiments. Elements and/or components in the figures are not necessarily drawn to scale. Throughout this disclosure, depending on the context, singular and plural terminology may be used interchangeably.

(2) FIG. 1A schematically depicts an upper perspective view of an exemplary multi-accessory mount assembly constructed in accordance with one or more embodiments of the disclosure.

(3) FIG. 1B illustrates a side perspective view of the multi-accessory mount assembly of FIG. 1A.

(4) FIGS. 1C to 1E illustrate the steps of coupling a sub mount to a proximal portion of the multi-accessory mount assembly in accordance with one or more embodiments of the disclosure.

(5) FIG. 2A illustrates an upper perspective view of another exemplary multi-accessory mount assembly constructed in accordance with one or more embodiments of the disclosure.

(6) FIG. 2B illustrates a side perspective view of the interchangeable sub mount of FIG. 2A.

(7) FIG. 3 illustrates a view of an exemplary multi-accessory mount assembly coupled to an exemplary optic in accordance with one or more embodiments of the disclosure.

(8) FIG. 4 illustrates a view of an exemplary multi-accessory mount assembly coupled to an exemplary optic in accordance with one or more embodiments of the disclosure.

(9) FIG. 5 illustrates a view of an exemplary one-piece multi-accessory mount assembly in accordance with one or more embodiments of the disclosure.

(10) FIG. 6A illustrates a side view of a multi-accessory mount assembly coupled to an optic and a laser in accordance with one or more embodiments of the disclosure.

(11) FIG. 6B illustrates a rear view of the multi-accessory mount assembly of FIG. 6A.

(12) FIG. 6C illustrates a side view of a multi-accessory mount assembly coupled to an optic and a

laser in accordance with one or more embodiments of the disclosure.

(13) FIG. 6D illustrates a rear view of the multi-accessory mount assembly of FIG. 6C.

(14) FIGS. 7A to 7E illustrate various views of another exemplary multi-accessory mount assembly coupled to another exemplary optic in accordance with one or more embodiments of the disclosure.

(15) FIGS. 7F to 7H illustrate the coupling mechanism of the multi-accessory mount assembly of FIGS. 7A-7E.

(16) FIGS. 8A to 8C illustrate various views of another exemplary multi-accessory mount assembly in accordance with one of more embodiments of the disclosure.

(17) FIG. 8D is a top view of the coupling mechanism of the multi-accessory mount assembly of FIGS. 8A to 8C.

(18) FIG. 9 is a flow chart illustrating the steps of an exemplary method for coupling a multi-accessory mount assembly to a firearm in accordance with one or more embodiments of the disclosure.

DETAILED DESCRIPTION

(19) Overview

(20) Described below are embodiments of a multi-accessory mount assembly that can be attached to a firearm. Methods of installing the multi-accessory mount assembly on the firearm are also disclosed. The firearm may include a barrel with a breech end and a muzzle end. In some instances, the firearm may be a rifle (e.g., fully automatic, semi-automatic or bolt-action), pistol, shotgun, or the like. In some instances, the firearm may be a rifle, such as an M-16 style rifle, an AR-15 style rifle, an AR-10 style rifle, or an M-4 style rifle, among others that include a Picatinny or Weaver rail positioned above the barrel. Any firearm may be used in association with the firearm sight disclosed herein.

(21) The multi-accessory mount assembly of the present disclosure solves several problems that military, law enforcement, first responders and civilians have encountered when using an IR laser on a rifle. A multi-accessory mount assembly in accordance with the present disclosure includes a base with an elevated mount, and sub mounts that attach to the base. For example, the sub mounts may be interchangeable. The multi-accessory mount assembly moves the position of the IR laser/illuminator to the approximate center of the rifle (e.g., approximately over the action), and elevates the IR laser/illuminator to avoid the IR emission from the IR laser/illuminator being blocked or obscured by the user's hand should the user hold the rifle with their support (or off) hand gripping the handguard. The interchangeable sub mount removably attaches to an optic, such as one or more of a scope, red dot, reflex sight, magnifier or the like. The interchangeable sub mount positions the optic so that it sufficiently clears the IR laser so as not to impede the operability of the optic. Because the sub mount is interchangeable, it may be sized for different optics, and/or for multiple components, such as a red dot/reflex sight and a magnifier. The sub mount attaches to the base in a fitted fashion so that the optic returns to zero if the sub mount is removed from the base and then subsequently reattached. The multi-accessory mount assembly frees up rail space on the firearm platform, balances the weight to center mass over the firearm, and provides a near co-witnessed and unobstructed path for the laser/illuminator to be mounted rearward of the support arm of the shooter/operator.

(22) Certain embodiments of the multi-accessory mount assembly are designed to be modular and perfectly complements superior shooting stances and body mechanics, while capitalizing on reduced fatigue (e.g., heads up versus hunched positions) while shooting, thus allowing proper body mechanics and positioning. The multi-accessory mount assembly further allows the operator to utilize modern and mission critical head-borne systems and personal protective equipment (PPE), e.g., helmets, night vision, gasmasks, communications equipment, body armor, face wraps, etc., and aids in shoulder transitions for offhand shooting and cross eye dominant individuals. The multi-accessory mount assembly is designed to be modular and scalable, and may accept multiple optics system configurations. The assembly further may include a space to mount a magnifier

behind the day optic.

(23) These and other embodiments of the disclosure will be described in more detail through reference to the accompanying drawings in the detailed description of the disclosure that follows. This brief introduction, including section titles and corresponding summaries, is provided for the reader's convenience and is not intended to limit the scope of the claims or the proceeding sections. Furthermore, the techniques described above and below may be implemented in a number of ways and in a number of contexts. Several example implementations and contexts are provided with reference to the following figures, as described below in more detail. However, the following implementations and contexts are but a few of many.

Illustrative Embodiments

(24) Referring now to FIGS. **1A** to **1E**, an exemplary multi-accessory mount assembly is provided. Multi-accessory mount assembly **10** may include proximal portion **12**, neck portion **14**, and distal portion **16**. Distal portion **16** extends from proximal portion **12** via neck portion **14**. Moreover, neck portion **14** may be angled such distal portion **16** is elevated higher than proximal portion **12**. As shown in FIG. **1A**, neck portion **14** may include cut outs **24** to reduce the weight of the multi-accessory mount assembly. In some embodiments, cut outs **24** may extend horizontally through neck portion **14**.

(25) Proximal portion **12** has lower surface **20** and upper surface **22**. Lower surface **20** is configured to be removably coupled to a firearm. In some instances, the firearm may be a conventional and/or tactical, and may be a rifle, pistol, shotgun or the like. However, the firearm may be any number of firearms, such as, but not limited to, an M-16 style rifle, an AR-15 style rifle, an AR-10 style rifle, or an M-4 style rifle, or the like with a rail system, such as a Picatinny rail or a Weaver rail. The firearm may generally include a barrel and a muzzle end. Proximal portion **12** may include one or more screws/knobs **28** for facilitating attachment of proximal portion **12** to the rail system of the firearm, e.g., the Picatinny rail or the Weaver rail.

(26) Distal portion **16** has an upper surface configured to be removably coupled to a firearm accessory, e.g., an IR laser/illuminator. The upper surface of distal portion **16** may include rail system **18** to facilitate coupling with the firearm accessory. For example, rail system **18** may include a Picatinny rail or a Weaver rail. Distal portion **16** may be elevated from the rail system of the firearm a predetermined height when proximal portion **12** is coupled to the rail system of the firearm, thereby creating a space sized and shaped to permit the support hand of the operator of the firearm to grip the handguard without interfering with the use of the firearm accessory.

(27) Upper surface **22** of proximal portion **12** may be configured to be removably coupled to an interchangeable sub mount, e.g., interchangeable sub mount **100**. For example, upper surface **22** may include one or more apertures **26**, e.g., threaded apertures, for receiving screws that pass through corresponding apertures in interchangeable sub mount **100** to tightly secure the interchangeable sub mount. For example, upper surface **22** may include a threaded aperture at a proximal region and another threaded aperture at a distal region. Accordingly, interchangeable sub mount **100** may include a corresponding number of apertures **106**, e.g., threaded apertures, such that apertures **106** may be aligned with apertures **26** of proximal portion **12**.

(28) As shown in FIG. **1C**, proximal portion **12** may include position reference features that ensure exact repositioning of interchangeable sub mount **100** relative to the proximal portion **12**. An example system of position reference features includes groove **32** sized and shaped to receive and mate with ridge **110** of interchangeable sub mount **100**. In addition, as shown in FIG. **1D**, interchangeable sub mount **100** may include groove **108** sized and shaped to receive and mate with protrusion **34** of proximal portion **12**. Accordingly, interchangeable sub mount **100** may be removably coupled to proximal portion **12** such that groove **32** mates with ridge **110**, groove **108** mates with protrusion **34**, and apertures **106** are aligned with apertures **26**, as shown in FIG. **1E**. Upon alignment, a fastener, e.g., screws **S**, may be used to couple interchangeable sub mount **100** with proximal portion **12** via apertures **26** and **106**, as shown in FIG. **1F**. Screws **S** may fit through

cutouts in the Picatinny rail, if necessary, which function as extensions of apertures **106** and **26**. With this mounting system for interchangeable sub mount **100**, interchangeable sub mount **100** can be repeatedly removed and re-installed, and return to zero each time.

(29) The upper surface of interchangeable sub mount **100** may be configured to be removably coupled to one or more firearm accessories, e.g., one or more of a red dot, reflex sight, scope, IR or thermal scope/monocular, or a magnifier. The term “optic” will be used herein to describe collectively an optic or an optic mount. The upper surface of interchangeable sub mount **100** may include a rail system **102** to facilitate coupling with the firearm accessory. For example, protrusions **102** may include a Picatinny rail or a Weaver rail. Moreover, interchangeable sub mount **100** may include opening **104** for weight reduction of interchangeable sub mount **100**. As shown in FIG. **1B**, height **H** of the upper surface of interchangeable sub mount **100** may be predetermined based on, for example, the desired height of the optic for the shooter and/or the attached optic so that it is operable over the firearm accessory attached to distal portion **16**. For example, some optics are shorter (e.g., distance from bottom surface of the optic to reticle) than other optics, and they may require a relatively taller interchangeable sub mount so the reticle of the optic is viewable on target over the firearm accessory.

(30) Referring now to FIGS. **2A** and **2B**, an alternative exemplary interchangeable sub mount for coupling with multi-accessory mount assembly **10** is provided. Interchangeable sub mount **200**, as shown in FIG. **2B**, may be constructed and coupled to proximal portion **12** similar to interchangeable sub mount **100**. For example, rail system **202** corresponds with rail system **102**, opening **204** corresponds with opening **104**, and one or more apertures **206** correspond with one or more apertures **106**. Interchangeable sub mount **200** differs from interchangeable sub mount **100** in that interchangeable sub mount **200** may have a longer longitudinal length than interchangeable sub mount **100**, and consequently can have two or more firearm accessories attached to it, such as a red dot/reflex sight and a magnifier or a thermal or IR scope or monocular. Accordingly, the operator of the firearm may select between interchangeable sub mounts **100** and **200** based on the desired firearm accessory(ies) to be coupled to the firearm.

(31) Referring now to FIG. **3**, coupling multi-accessory mount assembly **10** to an optic is described. Optic **300** may be removably coupled to an upper surface of the interchangeable mount, e.g., interchangeable sub mount **200**, such that optic **300** is positioned at a height **H**. Optic **300** may include base **302** and lens holder **306** for holding lens **308**. Moreover, optic **300** may include knob **304** which may be actuated to fix optic **300** to the interchangeable sub mount **100**.

(32) Referring now to FIG. **4**, coupling multi-accessory mount assembly **10** to optic **450** via interchangeable sub mount **400**, which is adjustable, is described. Interchangeable sub mount **400** may be constructed similar to interchangeable sub mounts **100** and **200** described above. For example, lower surface **404** of interchangeable sub mount **400** may be configured to be removably coupled to upper surface **22** of proximal portion **12**, e.g., via screw **S**, and upper surface **406** of interchangeable sub mount **400** may be configured to be removably coupled to a firearm accessory, e.g., optic **450**. Moreover, upper surface **406** may extend from lower surface **404** via middle portion **402**, which may have opening **408**. Unlike interchangeable sub mounts **100** and **200**, as shown in FIG. **4**, interchangeable sub mount **400** may have a parallelogram shape such that upper surface **406** is positioned distal to lower surface **404**. Accordingly, middle portion **402** may be angled.

(33) In addition, the lower portion of interchangeable sub mount **400** may include rail system **412**, which may be fixed to proximal portion **12**, such that middle portion **402** is slidably movable relative to rail system **410**. Accordingly, rail system **410** may include threaded apertures for aligning with the threaded apertures of proximal portion **12**, and for receiving screw **S**, such that rail system **410** may be fixed relative to proximal portion **12**. As shown in FIG. **4**, rail system **410** may include a plurality of openings **412** configured to engage with pins/knobs **414** (e.g., a screw), such that pins/knobs **414** may be disengaged with openings **412** to permit middle portion **402** to be moved along rail system **410** to a desired longitudinal position, and pins/knobs **414** may engage

with openings **412** to lock middle portion **402** in the desired longitudinal position relative to rail system **410** and proximal portion **12**.

(34) Optic **450** may be removably coupled to upper surface **406** of interchangeable sub mount **400**. Moreover, optic **450** may have one or more knobs for controlling operation of optic **450**. When multi-accessory mount assembly **10** is coupled to both optic **450** via interchangeable sub mount **400** and IR laser/illuminator **350**, optic **450** is raised by height H.

(35) FIG. 5 is a perspective view of an exemplary multi-accessory mount assembly where the sub mount is integrally formed with the proximal portion of the multi-accessory mount assembly, e.g., as a single piece. In such a configuration, sub mount **1000** portion is not interchangeable, but fixed with the rest of the multi-accessory mount assembly. This configuration also may allow for an optic to be above a laser such that the laser does not block the firearm operator's vision of the optic reticle on the target. Further, the one-piece design may provide stability and be configured to allow for mounting of a single type of optic. As shown in FIG. 5, upper portion **1006** of sub mount **1000** may extend from lower portion **1002** via middle portion **1004**. Middle portion **1004** may contain serrations on the front or rear. Moreover, sub mount **1000** may have a tapered shape such that the width of middle portion **1004** decreases from lower portion **1002** toward upper portion **1006**. For example, a proximal side of middle portion **1004** may extend vertically, whereas a distal side of middle portion **1004** may be angled.

(36) As shown in FIG. 5, multi-accessory mount assembly **10''** may be constructed similar to multi-accessory mount assembly **10**, with similar components having like prime reference numerals. For example, proximal portion **12''** corresponds with proximal portion **12**, neck portion **14''** corresponds with neck portion **14**, and distal portion **16''** corresponds with distal portion **16**. Multi-accessory mount assembly **10'** differs from multi-accessory mount assembly **10** in that multi-accessory mount assembly **10''** is one-piece with the proximal portion **12''**, neck portion **14''**, and distal portion **16''**, as described in further detail above.

(37) FIG. 6A is a side view of multi-accessory mount assembly **10** coupled to both optic **300** via interchangeable sub mount **100**, and to laser **350**. FIG. 6B is a rear view of multi-accessory mount assembly **10** coupled to optic **300** via interchangeable sub mount **100** and to laser **350**. Height H allows for optic **300** to be visible above laser **350**. As shown in FIG. 6B, lower surface **20** of proximal portion **12** may include channel **30** sized and shaped to engage with the rail mechanism of the firearm, e.g., at the middle region of the firearm (e.g., approximately over the action), via screws/knobs **28**. The line of sight through optic **300** may be above IR laser **350**, being raised by height H, such that IR laser **350** does not block the firearm operator's vision of the optic reticle on the target.

(38) FIG. 6C is a side view of multi-accessory mount assembly **10** coupled to both optic **450** via interchangeable sub mount **400**, and to laser **350**. As will be understood by a person having ordinary skill in the art, any of the firearm accessories described above may be interchangeably coupled to multi-accessory mount assembly **10**, and multi-accessory mount assembly **10** further may be removably coupled to other known firearm accessories. FIG. 6D is a rear view of multi-accessory mount assembly **10** coupled to optic **450** via interchangeable sub mount **400** and to laser **350**. The height H allows for optic **450** to be visible above the laser **350**. As shown in FIG. 6D, the reticle of optic **450** may be visible on target above IR laser/illuminator **350**, such that laser **350** does not block the firearm operator's vision of the optic reticle on the target.

(39) Referring now to FIGS. 7A to 7E, an alternative exemplary interchangeable sub mount for coupling with an alternative exemplary multi-accessory mount assembly is provided. Multi-accessory mount assembly **10'** may be constructed similar to multi-accessory mount assembly **10**, with similar components having like-prime reference numerals. For example, proximal portion **12** corresponds with proximal portion **12**, neck portion **14** corresponds with neck portion **14**, and distal portion **16** corresponds with distal portion **16**. Multi-accessory mount assembly **10'** differs from multi-accessory mount assembly **10** in that multi-accessory mount assembly **10'** includes groove

501 disposed between proximal portion 12 and neck portion 14 for releasably engaging with interchangeable sub mount 500, as described in further detail below. As shown in FIG. 7D, neck portion 14 may include cut outs 524 to reduce the weight of the multi-accessory mount assembly. In some embodiments, cut outs 524 may extend horizontally through neck portion 14.

(40) Interchangeable sub mount 500 may be constructed similar to interchangeable sub mounts 100, 200, and 400 described above. For example, lower portion 502 of interchangeable sub mount 500 may be configured to be removably coupled to proximal portion 12, and upper portion 506 of interchangeable sub mount 500 may be configured to be removably coupled to a firearm accessory, e.g., optic 450. Moreover, upper portion 506 may extend from lower portion 502 via middle portion 504. Middle portion 504 may contain serrations on the front or rear. Unlike interchangeable sub mounts 100, 200, and 400, as shown in FIG. 7A, interchangeable sub mount 500 may have a tapered shape such that the width of middle portion 504 decreases from lower portion 502 toward upper portion 506. For example, a proximal side of middle portion 504 may extend vertically, whereas a distal side of middle portion 504 may be angled.

(41) In addition, as shown in FIG. 7B, interchangeable sub mount 100 may include opening 512 extending from the proximal side of middle portion 504 to the distal side of middle portion 504 for weight reduction of interchangeable sub mount 500. An iron-sight system including bar 570 and ramp 580 may be disposed within opening 512. For example, the iron sight system may be configured such that ramp 580 is set within opening 512 and rear bar 570 is within cut out 24. As will be understood by a person having ordinary skill in the art, the iron-sight system may be any other iron sight system of one or two pieces, including but not limited to post systems, bead systems, ramp systems, ring systems, or combinations of these or other iron-sight systems.

(42) As shown in FIG. 7C, upper portion 506 may include one or more apertures 514, e.g., threaded apertures, for receiving screws that pass through corresponding apertures in the firearm accessory, e.g., optic 450, to tightly secure the firearm accessory to interchangeable sub mount 500. As shown in FIGS. 7A and 7D, interchangeable sub mount 100 may include openings 508 and 509, each extending laterally into a side of middle portion 504 for additional weight reduction of interchangeable sub mount 500. Openings 508 and 509 may be separated by a support-providing wall, such that neither openings 508 nor 509 extend completely through middle portion 504.

(43) Referring now to FIGS. 7E to 7H, coupling multi-accessory mount assembly 10' to optic 450 via interchangeable sub mount 500 is described. As shown in FIG. 7E, lower portion 502 may include screw bore 560 extending through a proximal portion of lower portion 502, and proximal portion 12 may have a corresponding screw bore 561 extending at least partially therethrough, which may be aligned with screw bore 560 to facilitate securement of interchangeable sub mount 500 to proximal portion 12, e.g., via a screw. As described above, multi-accessory mount assembly 10' includes groove 501 for releasably engaging with interchangeable sub mount 500. Accordingly, as shown in FIGS. 7G and 7H, lower portion 502 of interchangeable sub mount 500 includes ridge 510 sized and shaped to be securely received by groove 501. In addition, lower portion 502 of interchangeable sub mount 500 includes pin 516 extending outwardly from the bottom surface of lower portion 502, and multi-accessory mount assembly 10' includes receptacle 503, sized and shaped to securely and pivotally receive pin 516 therein.

(44) First, as shown in FIGS. 7G, interchangeable sub mount 500 may be positioned above and perpendicular to multi-accessory mount assembly 10', such that pin 516 is aligned with receptacle 503, and so that lowering of interchangeable sub mount 500 toward multi-accessory mount assembly 10' can be achieved without colliding ridge 510 and neck portion 14. When interchangeable sub mount 500 is in contact with multi-accessory mount assembly 10' such that pin 516 is received within receptacle 503, interchangeable sub mount 500 may then be rotated relative to multi-accessory mount assembly 10' as shown in FIG. 7H so that ridge 510 enters groove 501.

(45) FIG. 7D illustrates when interchangeable sub mount 500 is properly aligned with multi-accessory mount assembly 10', e.g., when lower portion 502 is parallel with proximal portion 12,

and screw bore **560** is aligned with screw bore **561**. A screw may then be inserted to secure interchangeable sub mount **500** to proximal portion **12**. The frictional forces between ridge **510** and groove **501**, the frictional forces between pin **516** and receptacle **503**, and the frictional forces between the bottom surface of lower portion **502** and the upper surface of proximal portion **12** may assist in securely maintaining interchangeable sub mount **500** in positioned relative to multi-accessory mount assembly **10'**.

(46) Referring now to FIGS. **8A** to **8D**, an alternative exemplary interchangeable sub mount for coupling with an alternative exemplary multi-accessory mount assembly is provided. FIG. **8A** shows a side view of a proximal portion **12** onto which the interchangeable sub mount **600** will be mounted. FIG. **8B** shows the bottom view of the proximal portion **12**, including showing screw bores extending therethrough. FIG. **8C** shows a cross-sectional view of the interchangeable sub mount **600**.

(47) As shown in FIG. **8A**, multi-accessory mount assembly **10'** may be constructed similar to multi-accessory mount assembly **10**, with similar components having like prime reference numerals. For example, proximal portion **12** corresponds with proximal portion **12**, neck portion **14** corresponds with neck portion **14**, and distal portion **16** corresponds with distal portion **16**. Multi-accessory mount assembly **10'** differs from multi-accessory mount assembly **10** in that multi-accessory mount assembly **10'** includes groove **601** disposed between proximal portion **12** and neck portion **14** for releasably engaging with interchangeable sub mount **600**, as described in further detail below.

(48) As shown in FIG. **8C**, interchangeable sub mount **600** may be constructed similar to interchangeable sub mounts **100**, **200**, **400**, and **500** described above. For example, lower portion **602** of interchangeable sub mount **600** may be configured to be removably coupled to proximal portion **12**, and upper portion **606** of interchangeable sub mount **600** may be configured to be removably coupled to a firearm accessory, e.g., optic **450**. Moreover, upper portion **606** may extend from lower portion **602** via middle portion **604**. For example, a proximal side of middle portion **604** may extend vertically, whereas at least a portion of a distal side of middle portion **604** may be angled. Middle portion **604** may contain serrations on the front or rear. In addition, interchangeable sub mount **600** may include opening **612** extending from the proximal side of middle portion **604** to the distal side of middle portion **604** for weight reduction of interchangeable sub mount **600**.

(49) Lower portion **602** may include one or more screw bores **660** extending through a proximal portion of lower portion **602**, and proximal portion **12** may have a corresponding screw bore **661** extending at least partially therethrough, which may be aligned with screw bore **660** to facilitate securement of interchangeable sub mount **600** to proximal portion **12**, e.g., via a screw. As shown in FIG. **8C**, proximal portion **12** may include two separate screw bores **660** for mounting interchangeable sub mount **600** thereupon, which may be aligned with the two screw bores **661** of proximal portion **12**, as shown in FIG. **8B**.

(50) FIG. **8D** shows the coupling mechanism of proximal portion **12** for securing interchangeable sub mount **600** to the proximal portion **12**. As described above, multi-accessory mount assembly **10'** includes groove **601** for releasably engaging with interchangeable sub mount **600**. Accordingly, groove **601** is sized and shaped to securely receive ridge **610** of lower portion **602** of interchangeable sub mount **600**, as shown in FIG. **8D**. In addition, proximal portion **12** includes receptacle **603** sized and shaped to securely and pivotally receive pin **616** extending outwardly from the bottom surface of lower portion **602** of interchangeable sub mount **600** therein.

(51) Referring now to FIG. **9**, exemplary method **700** for coupling multi-accessory mount assembly **10**, **10'** to a firearm is provided. At step **702**, proximal portion **12**, **12**, **12** of multi-accessory mount assembly **10**, **10'** may be coupled to the rail mechanism of the firearm, e.g., a Picatinny rail or a Weaver rail, via screws/knobs **28**. Multi-accessory mount assembly **10**, **10'**, **10''** should be coupled to the firearm such that distal portion **16**, is distal to proximal portion **12**, relative to the firearm. At step **704**, an interchangeable sub mount, e.g., interchangeable sub mounts **100**, **200**, **400**, **500**, **600**

may be selected by the firearm operator and coupled to proximal portion **12**, **12'**, **12''** using a system of position reference features depending on which interchangeable sub mount is selected, e.g., such that groove **32** mates with ridge **110**, groove **108** mates with protrusion **34**, apertures **106** are aligned with apertures **26**, pin **516** is received within receptacle **503**, ridge **510** enters groove **501**, screw bore **560** is aligned with screw bore **561**, pin **616** is received within receptacle **603**, ridge **610** enters groove **601**, and/or screw bore **660** is aligned with screw bore **661**, etc.

(52) Screws **S** may be used to fix the selected interchangeable sub mount to the proximal portion. For example, when interchangeable sub mount **500** is selected, interchangeable sub mount **500** may be coupled to proximal portion **12**, e.g., by engaging pin **516** and receptacle **503**, rotating interchangeable sub mount **500** until ridge **516** is engaged with groove **501**, and inserting screw **S** through screw bores **560** and **561**. When interchangeable sub mount **600** is selected, interchangeable sub mount **600** may be coupled to proximal portion **12**, e.g., by engaging pin **616** and receptacle **603**, rotating interchangeable sub mount **600** until ridge **616** is engaged with groove **601**, and inserting screws **S** through both screw bores **660** and both screw bores **661**. At step **706**, a first firearm accessory, e.g., laser **350**, may be coupled to the upper surface of distal portion **16**, **16'**, **16''**. At step **708**, a second fire arm, e.g., optic **450** or **300**, may be coupled to the upper surface of the selected interchangeable sub mount.

(53) Although specific embodiments of the disclosure have been described, numerous other modifications and alternative embodiments are within the scope of the disclosure. For example, any of the functionality described with respect to a particular device or component may be performed by another device or component. Further, while specific device characteristics have been described, embodiments of the disclosure may relate to numerous other device characteristics. Further, although embodiments have been described in language specific to structural features and/or methodological acts, it is to be understood that the disclosure is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as illustrative forms of implementing the embodiments. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments could include, while other embodiments may not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

Claims

1. A multi-accessory mount assembly, the multi-accessory mount assembly comprising: a proximal portion having (i) a base configured to be removably coupled to a firearm rail and (ii) a sub mount, wherein the base includes a bottom surface configured to engage the firearm rail, and the sub mount having a first upper surface configured to receive a first firearm accessory, wherein the first upper surface is substantially planar, and the first upper surface being a first height above the bottom surface; and a distal portion coupled to the base, wherein the distal portion having a second upper surface configured to receive a second firearm accessory, wherein the second upper surface is substantially planar, the second upper surface being a second height above the bottom surface; wherein the first height is different than the second height; wherein a distal end of the first upper surface and a proximal end of the second upper surface include a horizontal space there between along a longitudinal axis of the base.
2. The multi-accessory mount assembly of claim 1, wherein the sub mount is interchangeable.
3. The multi-accessory mount assembly of claim 1, wherein the proximal portion is configured to be removably coupled to the base.
4. The multi-accessory mount assembly of claim 1, wherein the proximal portion comprises a first threaded aperture and the sub mount comprises a second threaded aperture configured to be aligned

with the first threaded aperture, the multi-accessory mount assembly further comprising a screw configured to couple the sub mount to the proximal portion via the first and second threaded apertures.

5. The multi-accessory mount assembly of claim 1, wherein the sub mount comprises a first geometry configured to mate with a second geometry of the proximal portion.

6. The multi-accessory mount assembly of claim 5, wherein the first geometry comprises a pin and the second geometry comprises a receptacle configured to pivotally receive the pin, and wherein the sub mount further comprises a ridge configured to releasably engage with a groove on the base.

7. The multi-accessory mount assembly of claim 1, wherein, when the base is coupled to the firearm, a lower surface of the distal portion and an upper surface of the firearm opposite the lower surface define a gap therebetween.

8. The multi-accessory mount assembly of claim 1, wherein the second upper surface of comprises a rail system.

9. The multi-accessory mount assembly of claim 1, wherein the sub mount and the distal portion are configured to be coaxial along a length of the firearm when the multi-accessory mount assembly is mounted to the firearm.

10. The multi-accessory mount assembly of claim 1, wherein the sub mount extends in a substantially perpendicular direction from the proximal portion, and wherein the distal portion is parallel to the proximal portion and spaced apart from the firearm when coupled to the firearm.

11. The multi-accessory mount assembly of claim 1, wherein the first upper surface has a second longitudinal axis and the second upper surface has a third longitudinal axis, and wherein a first longitudinal axis and the second longitudinal axis are parallel.

12. The multi-accessory mount assembly of claim 1, wherein the first upper surface is configured to receive a sight.

13. A multi-accessory mount assembly, comprising: a base configured to be removably coupled to a firearm, wherein the base includes a lower portion that defines a channel configured to engage the firearm rail; a sub mount coupled the base, the sub mount having a first upper surface configured to receive a first firearm accessory, the first upper surface being a first height above the channel; and a distal portion couple to the base and extending from the base toward a muzzle end of the firearm and spaced apart from the firearm when mounted to the firearm to form a gap therebetween, the distal portion having a second upper surface configured to receive a second firearm accessory, the second upper surface being a second height above the channel, wherein the second upper surface comprises a mounting surface that is substantially planar; wherein the first height is greater than the second height.

14. The multi-accessory mount assembly of claim 13, wherein the sub mount includes a first opening in a longitudinal direction of the base.

15. The multi-accessory mount assembly of claim 14, wherein the distal portion includes a second opening longitudinally aligned with the first opening.

16. The multi-accessory mount assembly of claim 15, wherein the first opening and the second opening include an iron sight system.

17. The multi-accessory mount assembly of claim 13, wherein the second upper surface includes a picatinny rail system.

18. The multi-accessory mount assembly of claim 13, wherein the base is configured to attach to a picatinny rail of the firearm.
